

TASK #51637

Q Booting Process in Linux (Steps in detail), Important Port Numbers Some Advanced Server Administrator commands for troubleshooting and monitoring

1. Linux Booting Process

Step 1: BIOS/UEFI

- Computer powers on, BIOS/UEFI runs POST (Power-On Self-Test)
- Checks hardware (RAM, keyboard, disks)
- Finds the boot device and loads the bootloader

Step 2: Bootloader (GRUB)

- GRUB (GRand Unified Bootloader) loads
- Shows a menu (if multiple OS/kernels are installed)
- Loads the Linux kernel and initramfs into memory

Step 3: Kernel Initialization

- Kernel loads into memory and initializes
- Mounts the initramfs (temporary root filesystem)
- Detects hardware, loads necessary drivers/modules

Step 4: initramfs / initrd

- Temporary filesystem used to load drivers needed to mount the real root filesystem
- Once real root (/) is mounted, initramfs is unmounted

Step 5: init/systemd Process

- Kernel starts the first process — **systemd** (PID 1) on modern systems
- systemd reads its configuration and starts managing all other processes

Step 6: Runlevels / Targets

- systemd uses **targets** instead of old "runlevels"
- Common targets: multi-user.target

- **Step 7: Services Start**

- systemd starts services in order (networking, SSH, cron)
- Some run in parallel for faster boot

Step 8: Login Prompt

- User sees login screen (GUI) or terminal login prompt (CLI)
- System is now fully booted and ready

2. Important Port Numbers

<u>Port NO</u>	<u>Service</u>
20, 21	FTP (File Transfer)
22	SSH (Secure Shell)
23	Telnet
25	SMTP (Email sending)
53	DNS
67, 68	DHCP
80	HTTP (Web)
110	POP3 (Email retrieval)
123	NTP (Time sync)
143	IMAP (Email retrieval)
161	SNMP (Monitoring)
443	HTTPS (Secure Web)
3306	MySQL
3389	RDP (Remote Desktop)
5432	PostgreSQL
8080	HTTP Alternate (common for apps)

3. Advanced Server Admin Commands (Troubleshooting & Monitoring)

A) Checking System Resources

```
#top
```

Real-time process/CPU/memory usage

```
#htop
```

Better visual version of top

```
#free -h
```

Memory usage (human-readable)

```
#df -h
```

Disk space usage

```
#du -sh *
```

Size of files/folders in current directory

B) Checking Running Processes

```
#ps aux
```

List all running processes

```
#ps aux | grep nginx
```

Find specific process

```
#kill -9 (PID)
```

Force kill a process

```
#pkill -f "process_name"
```

Kill by name

C) Network Troubleshooting

```
#ping google.com
```

Check connectivity

```
#traceroute google.com
```

Trace network path

```
#netstat -tulnp
```

List listening ports

```
#ss -tulnp
```

Modern alternative to netstat

```
#nslookup google.com
```

DNS lookup

```
#dig google.com
```

Detailed DNS info

```
#curl -I https://example.com
```

Check HTTP headers/status

D) Log Monitoring

```
#tail -f /var/log/syslog
```

Live log monitoring

```
#journalctl -xe
```

Systemd logs (errors)

```
#journalctl -u ssh -f
```

Live logs for a specific service

```
#dmesg | tail
```

Kernel messages (hardware issues)

E) Service Management (systemd)

```
#systemctl status nginx
```

Check service status

```
#systemctl restart nginx
```

Restart a service

```
#systemctl enable nginx
```

Enable on boot

```
#systemctl list-units --failed
```

Show failed services

F) Disk & File System

```
#lsblk
```

List disks/partitions

```
#fdisk -l
```

Disk partition details

```
#mount /dev/sdb1 /mnt
```

Mount a partition

```
#fsck /dev/sda1
```

Check/repair filesystem

G) User/Security Auditing

```
#who
```

Currently logged-in users

```
#last
```

Login history

```
#sudo tail -f /var/log/auth.log
```

Auth attempts (SSH, sudo)

H) Performance Troubleshooting Combo (real-world example)

```
#top
```

Check CPU/memory hogs

```
#free -h
```

Check if RAM is full/swapping

```
#df -h
```

Check if disk is full

```
#iostat -x 1
```

Check disk I/O bottlenecks

```
#netstat -tulnp
```

Check for unusual open ports

```
#journalctl -p err -b
```

Check for boot-time errors